STPM 2025 PHYSICS 960/2 (U2)

Questions and Answers

Values of constants: $e = 1.60 \times 10^{-19}$ C, $m_e = 9.11 \times 10^{-31}$ kg, $k = \frac{1}{4\pi\varepsilon_0} = 9.0 \times 10^9\ N\,m^2\,C^{-2}$, $\varepsilon_0 = 8.85 \times 10^{-12}\ F\,m^{-1}$, $\mu_0 = 4\pi \times 10^{-7}\ H\,m^{-1}$.

Section A [15 marks]

1. Two spheres, S_1 and S_2, with charges $+Q$ and $-Q$ respectively lie in a straight line that are separated by a distance, r. A similar sphere, S_3, with charge $+Q$ is placed in the middle of S_1 and S_2. What are the magnitude and direction of the electrostatic force that acts on S_3?

A Zero

B $\frac{Q^2}{2\pi\varepsilon_0 r^2}$ towards $-Q$

C $\frac{Q^2}{\pi\varepsilon_0 r^2}$ towards $+Q$

D $\frac{2Q^2}{\pi\varepsilon_0 r^2}$ towards $-Q$

Answer: D

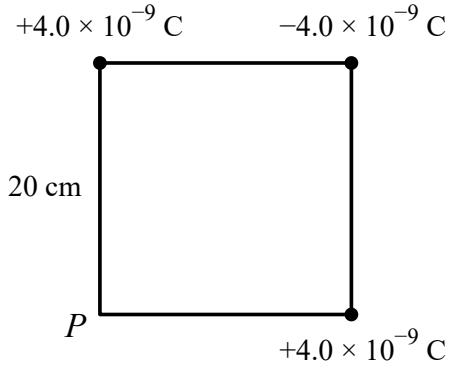
Distance from each charge to $S_3 = \frac{r}{2}$; each force $= \frac{Q^2}{4\pi\varepsilon_0 (r/2)^2} = \frac{Q^2}{\pi\varepsilon_0 r^2}$

Repulsion from $+Q$ and attraction to $-Q$ both point towards $-Q$, so they add:

$$F = \frac{2Q^2}{\pi\varepsilon_0 r^2} \text{ towards } -Q$$

Explanation: Halving the distance makes each force four times larger.

2. Three point charges, $+4.0 \times 10^{-9}$ C, -4.0×10^{-9} C and $+4.0 \times 10^{-9}$ C, are placed at three vertices of a square of sides 20 cm as shown in the diagram below.



What is the electric potential at P?

A 1.3×10^2 V B 1.8×10^2 V C 2.3×10^2 V D 4.9×10^2 V

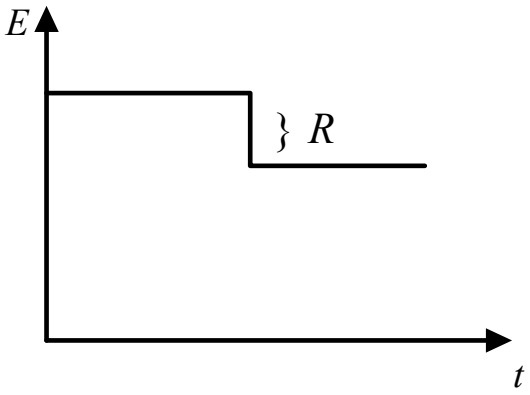
Answer: C

P is 0.20 m from each +4.0 nC charge and $0.20\sqrt{2} = 0.283$ m from the −4.0 nC charge.

$$V = (9.0 \times 10^9)(4.0 \times 10^{-9})\left(\frac{1}{0.20} + \frac{1}{0.20} - \frac{1}{0.283}\right) = 2.3 \times 10^2 \text{ V}$$

Explanation: Potential is a scalar: add the contributions with their signs.

3. The variation of electric field, E, across a parallel plate capacitor connected to a battery with time, t, is shown in the graph below.



Which statement is true about R?

A The capacitor starts to discharge.
B Another capacitor is connected in parallel to the capacitor.
C The distance between the plates of the capacitor is decreased.
D A dielectric is inserted between the plates of the capacitor.

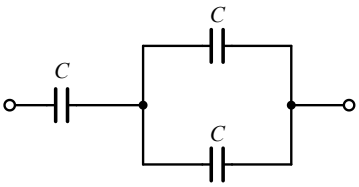
Answer: D

The field drops suddenly to a lower constant value at R. Inserting a dielectric reduces the field between the plates (D).

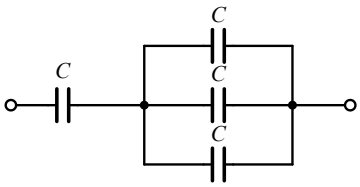
Explanation: As printed the question is ambiguous: with the battery still connected $E = V/d$ would stay the same. The drop fits a dielectric inserted into an isolated charged capacitor. Decreasing d would increase E; discharging gives a gradual decay; a parallel capacitor changes nothing.

4. Which arrangement of identical capacitors would give the effective capacitance of $\frac{1}{2}C$?

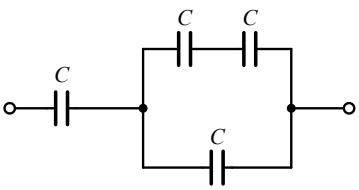
A



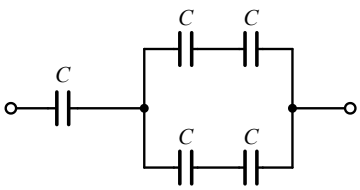
B



C



D



Answer: D

D: each branch is two C in series = $C/2$; the two branches in parallel = C.

$$\text{In series with } C\text{: } \frac{C \times C}{C + C} = \frac{1}{2}C$$

Explanation: A gives $\frac{2}{3}C$, B gives $\frac{3}{4}C$, C gives $\frac{3}{5}C$.

5. Which is true about drift velocity in a current-carrying conductor?

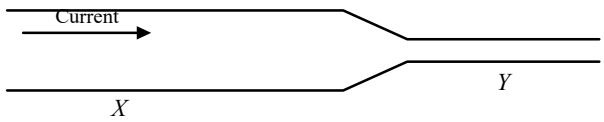
A The rate of change of electric current in the conductor.
B The instantaneous speed at which electrons move in the conductor.
C The average speed of electrons as the electrons travel in the conductor.
D The measure of the resistance experienced by electrons in the conductor.

Answer: C

Drift velocity is the average velocity of the free electrons along the conductor when a p.d. is applied (C).

Explanation: The instantaneous (random) speeds are much larger, about 10^6 m s^{-1}.

6. Two copper wires of different cross-sectional areas, X and Y, are joined together as shown in the diagram below.



Which statement is true about X and Y when a constant current flows through them?

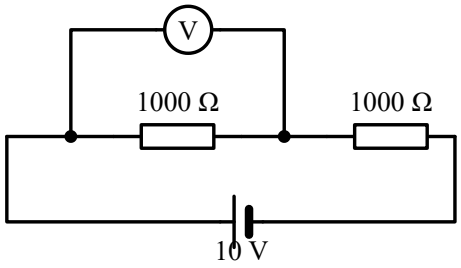
A The amount of work done to move a unit charge across X is less than that of Y.
B The rate of charge flows through a cross-sectional area of X is less than that of Y.
C The amount of work done to move a unit charge across X is greater than that of Y.
D The rate of charge flows through a cross-sectional area of X is greater than that of Y.

Answer: A

The same current flows through both. X is thicker, so its resistance per unit length is smaller and the p.d. (work per unit charge) across it is less (A).

Explanation: The rate of charge flow (current) is the same in X and Y.

7. Two identical resistors of resistance 1000 Ω are connected to a source of 10 V as shown in the circuit diagram below.



If a voltmeter of resistance 500 Ω is connected across one of the resistors, what is the reading of the voltmeter?

A 2.5 V B 3.3 V C 5.0 V D 7.5 V

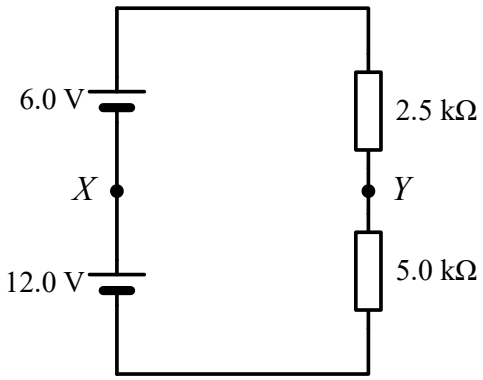
Answer: A

$$\text{Voltmeter} \parallel 1000\ \Omega: \frac{(500)(1000)}{500 + 1000} = 333\ \Omega$$

$$V = 10 \times \frac{333}{333 + 1000} = 2.5\ \text{V}$$

Explanation: A voltmeter with a low resistance changes the circuit and reads too low.

8. Two resistors of resistance, 2.5 kΩ and 5.0 kΩ, are connected to two batteries of 6.0 V and 12.0 V as shown in the circuit diagram below.



If a wire of resistance 5.0 Ω is connected to two points, *X* and *Y*, what is the current which flows in the wire?

A 0 A B 1.2 A C 2.4 A D 3.6 A

Answer: A

$$\text{Loop current} = \frac{6.0 + 12.0}{2.5 + 5.0}\ \text{mA} = 2.4\ \text{mA}$$

Potential of X above the bottom wire = 12.0 V (across the 12.0 V cell).

Potential of Y above the bottom wire = (2.4 mA)(5.0 kΩ) = 12.0 V

X and Y are at the same potential, so no current flows in the wire: 0 A.

Explanation: A balanced connection carries no current, whatever its resistance.

9. Two parallel wires each of length 3.0 m are separated by a distance of 5.0 mm. Each wire carries a direct current of 10.0 A in the opposite direction. What are the magnitude and direction of the force between both wires?

	Magnitude	*Direction*
A	1.2×10^{-5} N	Towards each other
B	1.2×10^{-5} N	Away from each other
C	1.2×10^{-2} N	Towards each other
D	1.2×10^{-2} N	Away from each other

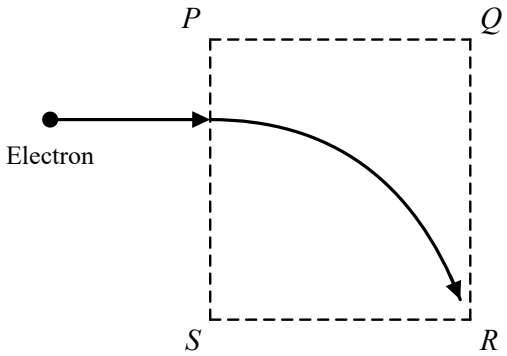
Answer: D

$$F = \frac{\mu_0 I^2 l}{2\pi d} = \frac{(4\pi \times 10^{-7})(10.0)^2(3.0)}{2\pi(5.0 \times 10^{-3})} = 1.2 \times 10^{-2}\ \text{N}$$

Opposite currents repel: away from each other.

Explanation: Like currents attract; unlike currents repel.

10. A moving electron enters perpendicularly through the side of PS to the page $PQRS$ and subsequently changes its direction to a parabolic path as shown in the diagram below.



Which statement is true about the page $PQRS$ that causes the change in the motion of the electron?

A A uniform electric field is directed from PQ to SR.
B A uniform magnetic field is directed parallel to SR.
C An electric potential decreases uniformly from SR to PQ.
D A uniform magnetic field is directed out of the page of $PQRS$.

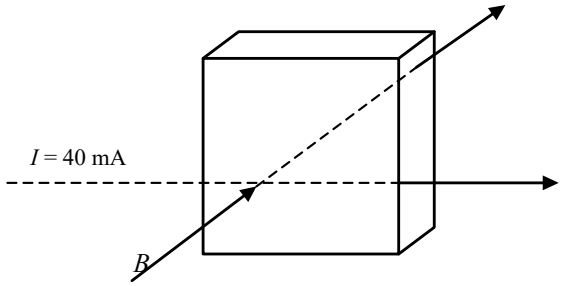
Answer: C

A parabolic path needs a constant force, i.e. a uniform electric field (a magnetic field gives a circle).

The electron is pushed towards *SR*, so the field points from *SR* to *PQ*; potential decreases along the field, from *SR* to *PQ* (C).

Explanation: The force on an electron is opposite to **E**, so A would bend it towards *PQ*.

11. A thin semiconductor plate of dimension 3.0 mm × 12 mm × 12 mm is placed at a right angle to a magnetic field, *B*, of 0.74 T as shown in the diagram below.



The current flows through the plate is 40 mA and the Hall voltage is 3.0 mV. What is the density of charge carriers of the semiconductor?

A $4.3 \times 10^{23}\ \text{m}^{-3}$ B $2.1 \times 10^{22}\ \text{m}^{-3}$ C $6.8 \times 10^{21}\ \text{m}^{-3}$ D $5.1 \times 10^{21}\ \text{m}^{-3}$

Answer: B

$$n = \frac{BI}{eV_H t} = \frac{(0.74)(40 \times 10^{-3})}{(1.60 \times 10^{-19})(3.0 \times 10^{-3})(3.0 \times 10^{-3})} = 2.1 \times 10^{22}\ \text{m}^{-3}$$

Explanation: *t* is the thickness of the plate along **B** (3.0 mm).

12. A pure inductor is connected to an a.c. supply of frequency 100 Hz and r.m.s. voltage of 12 V. If the peak current in the inductor is 14 A, what is the inductance of the inductor?

A 1.3 mH B 1.4 mH C 1.9 mH D 6.1 mH

Answer: C

$$X_L = \frac{V_0}{I_0} = \frac{12\sqrt{2}}{14} = 1.21\ \Omega$$

$$L = \frac{X_L}{2\pi f} = \frac{1.21}{2\pi(100)} = 1.9 \times 10^{-3}\ \text{H}$$

Explanation: Use peak voltage with peak current (or r.m.s. with r.m.s.).

13. A capacitor is connected to an a.c. supply of frequency 60 Hz. If the reactance of the capacitor is 390 Ω, what is the capacitance of the capacitor?

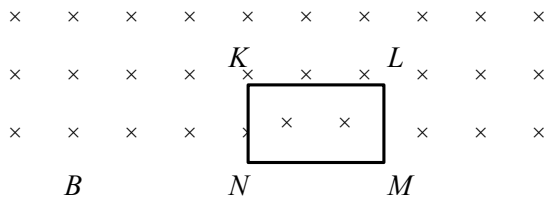
A 6.8×10^{-6} F B 2.1×10^{-5} F C 4.3×10^{-5} F D 2.6×10^{-3} F

Answer: A

$$C = \frac{1}{2\pi f X_C} = \frac{1}{2\pi(60)(390)} = 6.8 \times 10^{-6}\ \text{F}$$

Explanation: Reactance of a capacitor falls as frequency rises.

14. A loop, *KLMN*, is placed in a uniform magnetic field, *B*, as shown in the diagram below.



Which way would **not** affect the magnetic flux that passes through the loop?

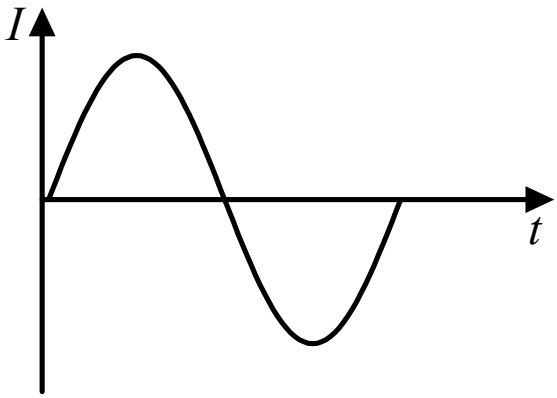
A By increasing the magnitude of *B*
B By moving the loop parallel to *B*
C By increasing the size of the loop
D By rotating the loop about an axis of *LM*

Answer: B

The field is uniform, so moving the loop parallel to **B** leaves *BA* unchanged (B).

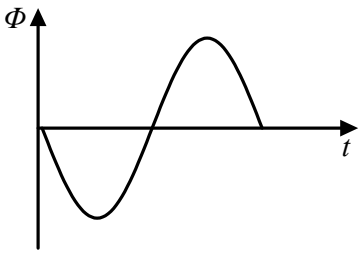
Explanation: Changing *B*, the area, or the angle (rotation about *LM*) all change $\Phi = BA\cos\theta$.

15. A variation of current, *I*, with respect to time, *t*, through an inductor is shown in the graph below.

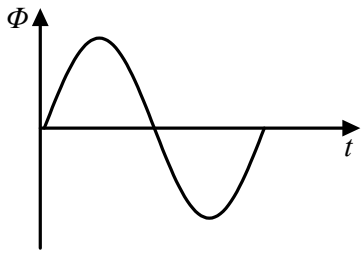


Which graph represents the variation of magnetic flux, Φ, with respect to time, *t*, through the inductor during the induction process?

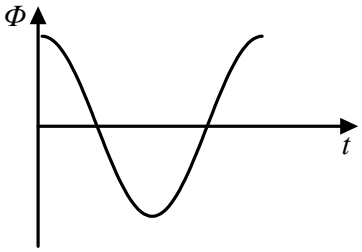
A



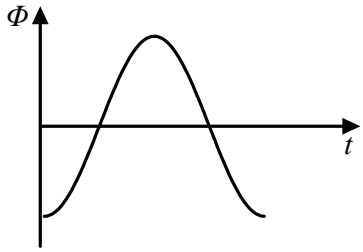
B



C



D



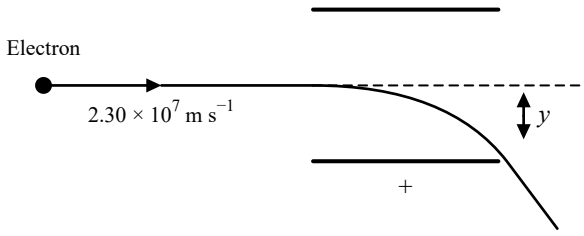
Answer: B

$N\Phi = LI$, so $\Phi \propto I$: the flux has exactly the same shape as the current (B)

Explanation: It is the induced e.m.f., $\propto$ dI/dt, that is shifted by a quarter cycle.

Section B [15 marks]

16. Two parallel metal plates of length 8.50 cm and separated by a distance of 2.50 cm are shown in the diagram below.



An electron travels with a speed of 2.30×10^7 m s^{-1} enters the region between the plates with right angle. The potential difference between the plates is 255 V.

(a) Calculate the time taken by the electron to move through the plates. [2]

Answer

$t = \frac{L}{v} = \frac{8.50 \times 10^{-2}}{2.30 \times 10^7}$ 1

$t = 3.70 \times 10^{-9}$ s 1

2

Explanation: The horizontal velocity is unchanged between the plates.

(b) Determine the vertical acceleration experienced by the electron. [4]

Answer

$E = \frac{V}{d} = \frac{255}{2.50 \times 10^{-2}} = 1.02 \times 10^{4}\ \text{V m}^{-1}$ 1

$F = eE = (1.60 \times 10^{-19})(1.02 \times 10^{4}) = 1.63 \times 10^{-15}\ \text{N}$ 1

$a = \frac{F}{m} = \frac{1.63 \times 10^{-15}}{9.11 \times 10^{-31}}$ 1

$a = 1.79 \times 10^{15}\ \text{m s}^{-2}$ (towards the positive plate) 1

4

Explanation: Gravity is negligible compared with the electric force.

(c) Determine the vertical deflection, y, for the electron on leaving the region. [2]

Answer

$y = \frac{1}{2}at^2 = \frac{1}{2}(1.79 \times 10^{15})(3.70 \times 10^{-9})^2$ 1

$y = 1.22 \times 10^{-2}\ \text{m}$ 1

2

Total for question **8**

Explanation: This is just under half the plate separation (1.25 cm), so the electron leaves the plates.

17. An alpha particle of charge, $+2e$, enters perpendicularly into a uniform magnetic field of 1.0 T at a constant speed of $7.0 \times 10^{6}\ \text{m s}^{-1}$.
[Mass of alpha particle is 6.64×10^{-27} kg.]

(a) Determine the radius of the path of the alpha particle. [4]

Answer

$qvB = \frac{mv^2}{r}$ 1

$r = \frac{mv}{qB}$ 1

$r = \frac{(6.64 \times 10^{-27})(7.0 \times 10^{6})}{2(1.60 \times 10^{-19})(1.0)}$ 1

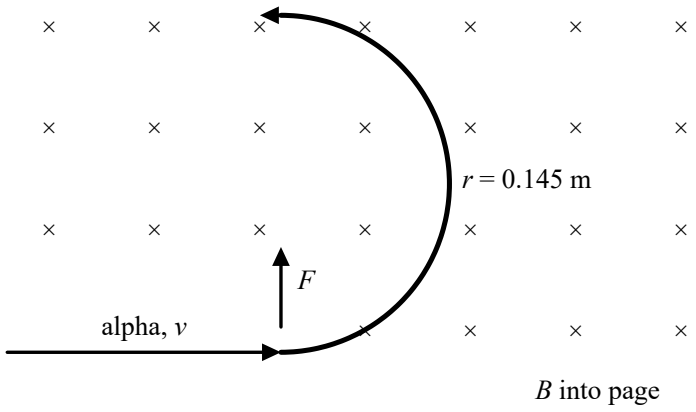
$r = 0.145\ \text{m}$ 1

4

Explanation: The charge of an alpha particle is $2e$.

(b) Sketch a labelled diagram to show the direction of the magnetic field and the path of the alpha particle. [2]

Answer



Magnetic field direction shown (e.g. into the page)	1
Circular path curving in the correct sense for a positive charge	1
	2

Explanation: Use Fleming's left-hand rule; the alpha particle is positive so the current is along **v**.

(c) Why does the speed of the alpha particle remains constant? [1]

Answer

The magnetic force is always perpendicular to the velocity, so it does no work and the kinetic energy (speed) is unchanged	1
	1
Total for question	**7**

Explanation: The force changes only the direction of motion.

Section C [30 marks]

18. (a) Define *capacitance* and *dielectric constant*. [2]

Answer

Capacitance: the ratio of the charge on either plate to the potential difference between the plates, $C = \frac{Q}{V}$	1
Dielectric constant: the ratio of the capacitance with the dielectric to the capacitance with vacuum (air), $\varepsilon_r = \frac{C}{C_0}$	1
	2

Explanation: Equivalently $\varepsilon_r = \varepsilon/\varepsilon_0$.

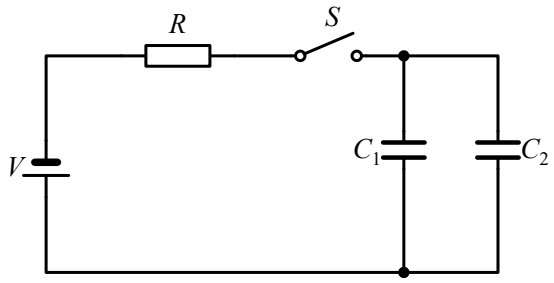
(b) A fully charged parallel plate capacitor has been disconnected from a battery. Explain the effect on the capacitance of the capacitor when a dielectric is inserted between the plates. [5]

Answer

The capacitor is isolated, so the charge Q on the plates stays constant	1
The dielectric molecules are polarised by the field between the plates	1
Induced surface charges on the dielectric produce a field opposing the applied field	1
The resultant field and hence the p.d. $V = Ed$ decrease	1
Since $C = \frac{Q}{V}$ with Q constant, the capacitance increases (by a factor ε_r)	1
	5

Explanation: The stored energy $Q^2/2C$ decreases.

(c) Two parallel plate capacitors, C_1 and C_2, are connected to a resistor, R, and a voltage supply, V, as shown in the circuit diagram below.



C_1 is filled with air and C_2 is filled with a dielectric of dielectric constant 3. Both capacitors have a plate area of 0.0050 m^2 and the distance between plates is 2.0 mm. The resistance of R is 20 kΩ. At time, $t = 0$, the switch, S, is closed for charging process.

(i) Determine the time when the current in the circuit becomes half of the maximum value. [6]

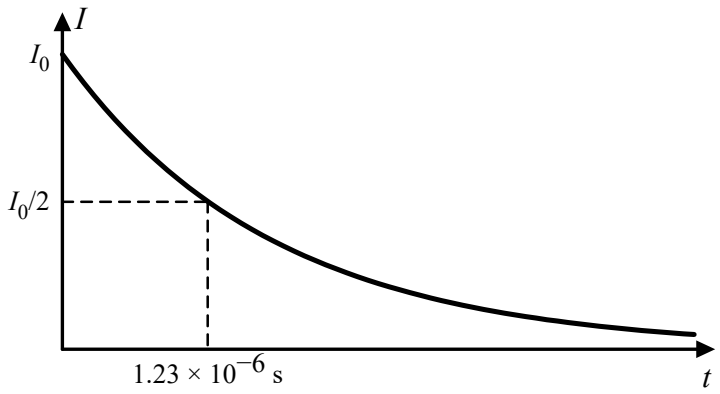
Answer

$C_1 = \frac{\varepsilon_0 A}{d} = \frac{(8.85 \times 10^{-12})(0.0050)}{2.0 \times 10^{-3}} = 2.21 \times 10^{-11}$ F	1
$C_2 = 3C_1 = 6.64 \times 10^{-11}$ F	1
In parallel: $C = C_1 + C_2 = 8.85 \times 10^{-11}$ F	1
$\tau = RC = (20 \times 10^3)(8.85 \times 10^{-11}) = 1.77 \times 10^{-6}$ s	1
$I = I_0 e^{-t/RC}$; $\frac{I_0}{2} = I_0 e^{-t/RC} \Longrightarrow t = RC \ln 2$	1
$t = (1.77 \times 10^{-6}) \ln 2 = 1.23 \times 10^{-6}$ s	1
	6

Explanation: The time for the current to halve does not depend on V.

(ii) Sketch a graph of current against time during the charging process. [2]

Answer



Exponential decay starting from a maximum value I_0 at $t = 0$	1
Approaching zero (asymptotically)	1
	2
Total for question	**15**

Explanation: The current is largest at the instant the switch is closed, when the capacitors are uncharged.

19. (a) Describe the motion of electrons in a conductor before and after a potential difference is applied. [4]

Answer

Before: free electrons move randomly at high speed in all directions, colliding with the lattice ions	1
Their average (net) velocity is zero, so there is no current	1
After: the electric field accelerates the electrons opposite to the field between collisions	1
A small drift velocity is superimposed on the random motion, giving a net flow of charge (current)	1
	4

Explanation: Drift velocities are of order 10^{-4} m s^{-1}.

(b) A copper wire of length 16 cm is connected to a potential difference of 14 mV.
[Number of free electrons per unit volume for copper is 8.5×10^{28} m^{-3} and resistivity of copper is 1.69×10^{-8} Ω m.]

(i) Determine the electric field in the wire. [2]

Answer

$E = \frac{V}{l} = \frac{14 \times 10^{-3}}{0.16}$ 1

$E = 0.0875$ V m^{-1} 1

2

Explanation: The field is uniform along a uniform wire.

(ii) Calculate the drift velocity of the free electrons in the wire. [5]

Answer

$J = \frac{E}{\rho}$ $(= \sigma E)$ 1

$J = \frac{0.0875}{1.69 \times 10^{-8}} = 5.18 \times 10^{6}$ A m^{-2} 1

$J = nev_d$ 1

$v_d = \frac{5.18 \times 10^{6}}{(8.5 \times 10^{28})(1.60 \times 10^{-19})}$ 1

$v_d = 3.81 \times 10^{-4}$ m s^{-1} 1

5

Explanation: The cross-sectional area is not needed: work with current density.

(c) Explain **two** effects on the resistivity if temperature increase for the following:

(i) Conductor [2]

Answer

The lattice ions vibrate with larger amplitude, so electrons collide more often (shorter mean free time); the number of free electrons is almost unchanged 1

Resistivity increases (roughly linearly with temperature) 1

2

Explanation: $\rho = m/(ne^2\tau)$: τ falls, n constant.

(ii) Semiconductor [2]

Answer

More electrons gain enough energy to cross into the conduction band, so the number of charge carriers (electrons and holes) increases greatly, outweighing the extra collisions	1
Resistivity decreases	1
	2
Total for question	**15**

Explanation: n rises much faster than τ falls.

20. (a) A solenoid of 200 turns with internal radius 2.0 cm and length 20 cm is connected to a battery of 12 V and a switch.

(i) How back e.m.f. is induced in the solenoid when the switch is closed? Explain your answer. [3]

Answer

When the switch is closed the current in the solenoid increases from zero	1
The magnetic flux linking the solenoid's own turns increases, so an e.m.f. is induced (Faraday's law)	1
By Lenz's law it opposes the increase in current, so it is a back e.m.f.	1
	3

Explanation: This is self-induction: $\varepsilon = -L\,\mathrm{d}I/\mathrm{d}t$.

(ii) Calculate the self-inductance of the solenoid. [3]

Answer

$L = \dfrac{\mu_0 N^2 A}{l}$	1
$L = \dfrac{(4\pi \times 10^{-7})(200)^2 \pi (0.020)^2}{0.20}$	1
$L = 3.16 \times 10^{-4}$ H	1
	3

Explanation: Use the radius 2.0 cm for the cross-sectional area.

(iii) Calculate the rate of the change of current at the instant the switch is closed. [2]

Answer

At $t = 0$, $I = 0$ so $\varepsilon = L\frac{dI}{dt}$: $\frac{dI}{dt} = \frac{12}{3.16 \times 10^{-4}}$ 1

$= 3.8 \times 10^{4}$ A s^{-1} 1

2

Explanation: At the instant of closing, the whole battery e.m.f. is balanced by the back e.m.f.

(b) A generator consist of N turns of coil with an area of 2.6×10^{-2} m^2 that rotates at the frequency of 90 Hz in a uniform magnetic field of 0.30 T. The maximum induced e.m.f produced is 22.1 V.

(i) Determine N of the generator. [5]

Answer

$\varepsilon_0 = NBA\omega$ 1

$\omega = 2\pi f = 2\pi(90) = 565$ rad s^{-1} 1

$N = \frac{\varepsilon_0}{BA\omega}$ 1

$N = \frac{22.1}{(0.30)(2.6 \times 10^{-2})(565)} = 5.01$ 1

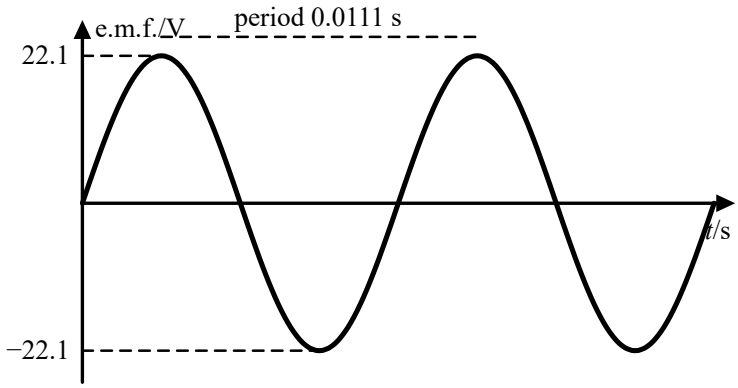
$N = 5$ 1

5

Explanation: The number of turns must be a whole number.

(ii) Sketch a graph of induced e.m.f. against time for the generator. [2]

Answer



Sinusoidal graph	1
Peak 22.1 V and period 0.0111 s marked	1
	2
Total for question	**15**

Explanation: $\varepsilon = \varepsilon_0 \sin \omega t$ when the coil starts with its plane perpendicular to **B**.